Assessing Compatibility of Natural Gas Pipeline Materials with Hydrogen, CO_2, and Ammonia

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Abstract: In this study, we examine the efficacy of repurposing natural gas (NG) pipelines for transporting hydrogen blends (with NG), ammonia, and CO_2 (gaseous and supercritical) from the standpoint of materials compatibility. Some information pertaining to component performance is also included, especially those components critical for pressurization and monitoring of flow. A listing of critical pipeline components and materials was developed, and their compatibilities was assessed for each fluid or gas type based on known compatibilities. Results indicate that pipeline materials should be suitable for gaseous CO_2 and anhydrous ammonia, but hydrogen blends greater than 12% may be problematic. Current compressor/regulator stations will not be suitable for use with either supercritical CO_2 or ammonia. Important knowledge gaps were identified, including (1) polymer performance with hydrogen/NG blends at low pressures, (2) compressor/regulator station polymers and epoxy coating materials with supercritical CO_2, and (3) metal performances of hydrogen/NG blends at low pressures.
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Introduction

Background

Natural gas (NG) pipeline systems consist of transmission lines, distribution lines, and compressor/regulator stations with approximately 3 million miles of pipelines used to transport and distribute natural gas in the United States (USEIA 2022). This extensive infrastructure was largely built over a period of 80 years, spans the continental United States, and represents a massive investment that supports the delivery of energy across the nation. As our energy systems transition from using traditional hydrocarbon resources to more renewable and nuclear sources there will likely be a need to transport other fluids at scales approaching our current natural gas system both to reduce atmospheric levels of carbon and as alternative energy carriers. Repurposing our existing NG pipeline system offers a potentially less expensive and more rapidly deployed approach for transporting large volumes of hydrogen (as blends with NG), ammonia, or CO_2 (supercritical and gaseous), but the feasibility of using the system to transport alternative fluids requires assessing the compatibility of the materials in the system with the alternative fluids.

A listing of the structures and relevant properties of hydrogen, CO_2, and ammonia is shown in Fig. 1 along with methane (the primary component of natural gas). Carbon dioxide, ammonia, and methane are relatively small molecules, but, unlike hydrogen, they are too large to diffuse through metals. However, permeation does occur with many polymers. Here it is a function of molecular size (diffusivity) and the van der Waals forces (which includes polar contributions) between the fluid molecule and the solid polymer. As shown in the table, hydrogen and methane are nonpolar, while both CO_2 and ammonia are polar. Water solubility is another important characteristic since water is a major contaminant in pipelines and the dissolution of chemical species (into water) can contribute to aqueous corrosion in steels if acids are formed. Both hydrogen and methane are relatively insoluble in water, CO_2 is moderately soluble, and ammonia is highly soluble. The dissolution of CO_2 into water will lower the pH due to the formation of carbonic acid, while ammonia, in marked contrast, will raise the pH.

Approximately 300,000 mi of transmission lines are used to move NG over long distances (Schmura et al. 2005). These pipes can have diameters up to 1.2 m (48 in.) and are composed almost entirely of low-carbon steels, though some cast, ductile, and wrought irons are still in use. The gauge pressure in these systems can range from 1.4 to 10.3 MPa (200 psi to 1,500 psi) (Schmura et al. 2005). For most transmission systems the line pressure is relatively constant; however, some pipeline companies do not have separate storage facilities and use the transmission piping itself for overnight storage. For these systems the pressure can increase by several MPa (100 psi) during low usage. Distribution lines transport the natural gas at gauge pressures <0.7 MPa (<100 psi) to homes and residential use. These pipes are smaller in diameter and are typically composed of either steel or plastic (polyethylene, polyvinyl chloride, or polyamide). Because of its lower cost and versatility, polyethylene is becoming more predominant in distribution systems. In addition to the pipes themselves, existing NG systems include compressor stations which contain, not only compressors, but also regulators, meters, valves, and other components composed of various polymers and both ferrous and nonferrous metals. These subunits of the compressor station are also expected to be in contact with, and therefore compatible with, the flowing gas or fluid. There are also regulators and meters to

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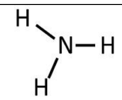
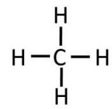
Gas/fluid Type	Structure	Polarity	Solubility in water at 20°C and 1 atm (g/kg_{water})
Hydrogen	H—H	Nonpolar	0.0016
Carbon dioxide	O=C=O	Polar	1.69
Ammonia		Highly polar	529.0
Methane		Nonpolar	0.023

Fig. 1. The molecular structure, polarity, and water solubility (Engineering ToolBox 2008) of hydrogen, carbon dioxide, ammonia, and methane.

reduce and monitor pressure from the transmission to distribution lines for residential use.

The existing pipeline infrastructure provides an established and low-cost option for moving hydrogen, ammonia and/or CO_2, around the country to advance hydrogen use and carbon sequestration. Most of this interest has centered on hydrogen since the use of existing infrastructure facilitates the transition to a hydrogen-based economy (Bjorck and Grahn 1999; Melaina et al. 2013; Goodhand et al. 2021; Tzimas et al. 2007; Clees et al. 2021; Pellegrini et al. 2020; Cerniauskas et al. 2020; Dodds and Demoullin 2013; Dickinson et al. 2010; Obara and Li 2020). While it is not quite feasible to use the existing infrastructure to move 100% hydrogen (since it is beyond the design capabilities of existing compressors), the ability to introduce hydrogen as a low-level blend with natural gas is possible. In fact, some small percentage of hydrogen (<1%) is present in some systems as a byproduct of coal gasification or from synthesis of methane from naphthalene. For the past 50 years, HawaiiGas has been providing synthetic natural gas containing 10%–12% hydrogen to its customers in Oahu (Hawai'iGas 2022).

As a result of the gathering interest in using the natural gas network to transport hydrogen as blends with natural gas, pipeline operators are conducting investigations of the resiliency of their systems with hydrogen/NG blends. Early off-the-record studies indicate that NG blends with up to 8% hydrogen can be handled in the existing infrastructure without any concerns. A more detailed discussion of hydrogen blend surveys is conducted in the Metallic Pipeline Materials section of this paper.

Ammonia has also received considerable attention in recent years as a potential hydrogen carrier (Wan et al. 2021). In fact, liquid ammonia has a higher concentration of hydrogen atoms per volume than does liquid hydrogen. The ammonia molecule can either be combusted directly in air or it can be dissociated at the end site to release the bound hydrogen atoms. In spite of the additional energy needed to crack ammonia to release hydrogen, it has an advantage of not promoting embrittlement, having higher hydrogen density than neat hydrogen and is less flammable. However, it is toxic and has known compatibility concerns, especially when water is present. Ammonia is extensively used in agriculture, and an existing piping network exists specifically for transporting and storing of anhydrous ammonia. It is worth noting that the materials used in these systems are not necessarily the same as those for NG use.

Another potential use of the natural gas pipelines is to transport CO_2 that has been captured and collected from point sources for enhanced oil recovery and for future direct air capture facilities for subsequent sequestration at a single-point location. Using the existing NG pipeline infrastructure to move captured CO_2 is a low-cost option compared to other transportation methods. Currently, CO_2 is used in the oil and gas industry to enhance the collection of crude oil. The transport piping is specifically designed for moving supercritical CO_2 (sCO_2).

The repurposing of portions or all of the NG infrastructure for hydrogen, ammonia, or CO_2 transport is attractive since it reduces or eliminates the costs and time to design, procure, and install new piping. However, before natural gas piping can be repurposed, it is important to determine whether any compatibility issues exist and where they are located. In this study, we explore the known material compatibility concerns with hydrogen, ammonia and sCO_2 with natural gas piping components. A listing of suitability was determined as a function of piping material and for the components in compressor and regulator stations. It is also worth noting that sCO_2 and ammonia will need to be prevented from entering distribution lines which are connected to homes, offices, etc. Therefore, only transmission pipelines are considered for repurposing to CO_2 and ammonia.

Hydrogen as a Fuel

Hydrogen has been intensely studied as a replacement for petroleum fuel for the past 40 years (Abe et al. 2019). Its key attractive feature is that it does not contain carbon atoms and therefore does not produce harmful CO_2 and CO emissions when reacted with oxygen to produce heat or electrical power. When operated in a fuel cell, only water is formed by the reaction chemistry; however, when burned in an engine, the thermal energy will cause atmospheric nitrogen to oxidize into harmful NOx emissions. It is important to note that NOx emissions emanating from internal combustion engines are routinely abated using NOx absorber technology (Gu and Epling 2019). An economy powered by hydrogen would reduce harmful greenhouse gas (GHG) emissions and reduce reliance on fossil fuels. However, few materials are suitable for storing and transporting hydrogen since its small size allows it to diffuse rapidly through many polymers and metals.

Hydrogen storage and containment is challenging and, due to the costs associated with liquification, hydrogen is primarily stored and transported as a gas. For pressures above 29.9 MPa (4,340 psi), the preferred tank materials for hydrogen storage are carbon fiber or fiberglass reinforced plastics with polymer liners to minimize tank

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mass while meeting pressure specifications (Type IV and Type V tanks). Lower pressure storage containers, up to 20 MPa (2,900 psi), and pipelines typically utilize metallic materials. Many metals, especially ferrous alloys, are prone to hydrogen embrittlement, therefore many alloys are not considered suitable for use. This effect is especially true for many high strength and high hardness steels subjected to high stress or fatigue levels in the presence of elemental hydrogen. These steels are vulnerable to catastrophic brittle fractures, even at service stresses below their stated yield strength.

Hydrogen embrittlement of steels (including pipeline steel grades) has been intensely studied to determine the underlying mechanisms and limits governing crack growth caused by hydrogen permeation (Baek et al. 2014; Briottet et al. 2012; Drexler et al. 2017; Ez-Zaki et al. 2020; Fassina et al. 2013; Fekete et al. 2015; Meng et al. 2017; Nguyen et al. 2020a, b; Nguyen et al. 2021; Nyukyforchyn et al. 2010; Ogawa et al. 2017, 2018; Pan et al. 2012; Ronevich et al. 2020; Sarzosa et al. 2020; Stalheim et al. 2014; Zhai et al. 2016). Hydrogen concentration and pressure play a role in embrittlement, but the steel grade and the associated mechanical properties of the steel and the welds also influence the embrittlement phenomenon.

Hydrogen embrittlement is potentially a very serious concern for existing natural gas pipelines if switching to carry pure hydrogen or blends containing a high percentage of hydrogen. When carbon or low alloy steels are exposed to hydrogen, the hydrogen will diffuse into the metal where it can be trapped at microstructural defects such as voids, grain boundaries, dislocations, and precipitates as well as regions of elevated strain (or stress). These trap sites enriched with hydrogen become active sites for embrittlement. The primary degradation mechanisms are fatigue crack growth under cyclic loading and slow crack growth under stable loads particularly near welds and other heat affected zones in pipes. Reports indicate that in a hydrogen atmosphere, rates of fatigue crack growth can be at least doubled, while crack propagation can be initiated at much smaller flaws (An et al. 2017; Amaro et al. 2014; Michler et al. 2021; San Marchi and Ronevich 2021). Steels experiencing hydrogen-initiated degradation typically fail in a brittle mode rather than the normal ductile mode with significant plastic deformation (Nguyen et al. 2020a, b). Sources of mechanical loading include residual stresses, applied internal load from the pressure of the gas being transported, operational fatigue from pressure fluctuations, static load from steady state operations, and plastic deformation during installation or at dents.

Formal recommendations or guidelines do exist that suggest limits on the grade or mechanical properties of pipeline steels for hydrogen service. ASME B31.12 recommends a maximum grade of X52 while API guidelines call for a maximum yield strength of 827 MPa (120 ksi) (ASME 2019; API 2020). Other recommendations include a maximum Vickers hardness of 235, particularly for heat affected zones, as well as the actual longitudinal, girth and repair welds (Goodhand et al. 2021). Several studies are keen to emphasize that the acceptable flaw size which can be tolerated is much smaller in pipelines transporting hydrogen as compared to natural gas transport (Goodhand et al. 2021; Pluvinage et al. 2019). As a result, more frequent pipeline inspections would likely be needed (Pluvinage et al. 2019).

Several studies suggest that blends containing up to 17% hydrogen can be successfully transported through existing natural gas pipelines (Haeseldonckx and D'Haeseleer 2007). However, the acceptable blend concentration will likely vary significantly between pipeline systems and hydrogen-natural gas compositions, and each system and gas mixture will need to be evaluated independently. The blending level, particularly if >20 vol%, can also impact the performance of appliances and other systems which require a pre-determined ratio of fuel and air with fuel blends containing higher proportions of hydrogen, which can cause problems (DeVries et al. 2007, 2017; Ishimoto et al. 2020; Jaffe et al. 2017; Melaina et al. 2013). This concern is driven by the knowledge that the gas flow and combustion behaviors are appreciably altered when hydrogen is added to natural gas (Florisson 2014). The US Department of Energy is conducting ongoing studies to evaluate metals and polymers under different hydrogen/natural gas blend ratios and at pressures up to 10 MPa (HYBLEND 2021). These efforts have primarily focused on the structural and mechanical properties of exposed materials.

Ammonia as a Hydrogen Carrier

Ammonia is also being studied as a hydrogen carrier (Cardoso et al. 2021; Ishimoto et al. 2020). It has the advantage of having a higher hydrogen content than an equivalent volume of hydrogen gas or liquid (at the same pressure). Ammonia also does not contain the flammability and explosiveness of hydrogen. From a materials compatibility standpoint, it also does not cause metal embrittlement since the hydrogen atoms are bound in a larger-sized molecule. Ammonia is produced at large scales to serve the agriculture industry as a fertilizer. As such, there exists extensive knowledge and experience associated with handling and the transport and storage of ammonia.

Carbon Dioxide Mitigation

CO_2 is a product of combustion of hydrocarbon fuels and is a strong greenhouse gas (GHG). Over the past 20 years, efforts have been intensified to reduce CO_2 emissions by utilizing fuel chemistries that do not contain carbon atoms (Cardoso et al. 2021). These have focused on eliminating CO_2 emissions from combustion primarily through the use of hydrogen (and to a much lesser extent ammonia), and/or removal and capture of CO_2 from the exhaust. Any captured CO_2 must be subsequently sequestered (fixated) to prevent further release to the atmosphere. Sequestration at the point of generation is typically not feasible for large volume sources since large scale CO_2 sequestration is expected to occur predominantly in suitable geological repositories with low regional density. As a result, there is interest in potentially repurposing existing natural gas pipelines to transport CO_2 to be sequestered at locations capable of handling large quantities of CO_2 (Blackburn 2022; Wilson 2021). In the United States, over 5,000 mi of pipelines exist for large-scale CO_2 transport. Currently, the major source (~80%) of CO_2 transported through pipelines in the US is reported to be from natural or geologic sources (underground CO_2 reservoirs) while combustion or processing of fossil fuels provides the remainder (Wallace et al. 2015). For these systems, CO_2 is primarily used to enhance tertiary oil production from subsurface oil reservoirs. These commercial CO_2 pipelines normally operate at gauge pressures between 8.3–15 MPa (1,200–2,200 psi), which places CO_2 in a dense supercritical state (where it exhibits both liquid and gas properties). Carbon dioxide becomes supercritical above 304.13K (31.0 °C or 87.8 °F) and 7.377 MPa (73.8 bar or 1,070 psi).

Pure, impurity-free CO_2 is not considered corrosive to commonly used pipeline steels. However, in its supercritical state, CO_2 is a proven solvent for many polymers. In addition, both naturally occurring CO_2 and the effluent from most combustion processes are expected to contain impurities like H_2O, O_2, H_2S, SO_2, NO_X, and NH_3. Consequently, most studies on supercritical CO_2 (sCO_2) reported in the literature address the effect of individual and mixed impurities (Wetenhall et al. 2014).

The removal of water is important in these systems since aqueous condensate will dissolve CO_2 to form corrosive carbonic acid. Furthermore, the physical properties of CO_2 differ substantially from natural gas as indicated in Fig. 1. As a result, the equipment used to pressurize and move CO_2 in these pipes has different operating parameters than those used in NG applications.

Natural Gas Pipeline Systems and Their Materials

The main elements of the natural gas distribution system include piping segments, compressor stations, and regulator stations. Compressor stations may both pressurize and purify the gas to maintain flow and gas quality while regulator stations typically reduce pressure between transmission and distribution systems. Material selection for pipelines is based on a combination of available strength, durability, and cost requirements. The economic opportunities for reducing cost have generally been held in check in recent years due to increasing safety and environmental regulatory concerns. Since most containment materials are compatible with natural gas, selection is primarily based on stress and strength requirements necessary to maintain the transport pressure under safe conditions. High strength, low carbon steels are used exclusively for the larger diameter (50–122 cm) high-capacity transmission lines since the pressures in these systems can be up to 10.3 MPa (1,500 psi). Gas pressure is regulated below 0.69 MPa (<100 psi) for end point distribution to residences and businesses. These pipes have much smaller diameters and can be either steel or plastic. Plastic is becoming the predominant pipe material for the low-pressure natural gas distribution lines because of its lower capital cost and easier installation.

To maintain proper natural gas flow in transmission line piping, periodic boosting of the pressure is necessary. This is achieved via compressor stations which are located approximately every 65 to 113 km (40 to 70 mi) of transmission pipe length. Compressor stations perform three primary functions: (1) purify the gas entering the compressor facility, (2) compress (or pressurize) the gas, and (3) cool the hot pressurized gases back to ambient conditions before exiting the station. Regulator stations are also employed to reduce and regulate the gas pressure to usable levels for the end user.

For each compressor and regulator station, there is an assortment of meters, regulators, filters, valves, sensors, and fittings to ensure performance and provide emergency shutoff. The range of materials used in these sub-components is much larger than those used for pipe construction. The compatibility of these materials with the gas chemistry is critical to ensure safe and proper operation. A key objective of this study is to identify those materials endemic to compressor/regulator stations and assess their compatibility with hydrogen/natural gas blends, ammonia, and CO_2.

Metallic Pipeline Materials

Pipeline steels for NG transport are expected to follow the American Petroleum Institute (API) standard rating system and are low carbon steels designated as grades that include A25, A, B, X42, X52, X60, XC60, X70, X80, and X100. The number following X represents the minimum yield strength of the steel in units of kilopounds per square inch (Nguyen 2021). These pipes are designated as seamless, welded without filler metal, or welded with filler metal according to their fabrication method. They are sold as either as-rolled or heat treated (to remove residual stresses). The safety factors (SFs) are based on the minimum yield strength and can range from 1.25 to 2.5 depending on application and location. For instance, pipelines buried near schools or residential areas will have a higher SF than pipes located in remote, less populated areas. Many of these pipelines have an interior epoxy coating to provide a smooth inner surface to minimize deposit build-up, corrosion product accumulation, and wall friction, which in turn, reduces pumping losses (Collet and Chizat 2015).

Hydrogen and Its Blends with Natural Gas

Hydrogen atoms readily diffuse through metals, especially steels. This infusion of hydrogen atoms into the steel structure will impact the mechanical and physical properties of the metal. One well-known consequence is embrittlement of the steel (increasing the hardness and reducing the plasticity). This phenomenon increases the crack growth rate and/or reduces the fatigue life of the metal structure. Hydrogen infusion and its subsequent embrittlement of metals is rapid and reversible (Dear and Skinner 2017). This effect is especially problematic for high strength steel grades, which are inherently less ductile and more prone to rapid crack growth than lower strength grades. As a result, many piping system design standards recommend lower strength steels (e.g., X42 and X52) for use with hydrogen. Embrittlement increases with hydrogen concentration and pressure. Most studies that have examined hydrogen embrittlement in API steels have been concerned with microstructural or property effects, especially fatigue crack growth, of a particular steel grade with hydrogen content and pressure. The internal epoxy coatings used to reduce gas-wall friction are not expected to impede hydrogen diffusion from the moving gas to the steel structures. Other commonly used metals, such as copper and aluminum, are less susceptible to hydrogen embrittlement.

Because hydrogen permeation also increases the yield strength of steel, in addition to increasing the hardness, the API safety factors have no relevance since the failure mode (fracture) does not correlate with yield strength. Therefore, other failure criteria associated with fracture toughness, such as fatigue, and elongation (or strain) to failure are much more applicable. To date, there has been no effort to assign an acceptable limit of fracture toughness, fatigue life, or strain to failure to assess suitability of pipeline steel with hydrogen content and pressure.

Stresses can be produced from the internal gas pressure, external loads such as constrained thermal expansion resulting from heating or cooling due to diurnal ambient temperature changes, or external pressures associated with ground water. The heat affected zones of welds are particularly vulnerable to hydrogen embrittlement because they can be subject to residual stresses that are significantly larger than the stress levels of the main pipe body.

In general, low yield strength steels, X52 or less, are considered suitable for use with hydrogen blends up to 12% (based on the long-term usage of 10%–12% hydrogen blends by HawaiiGas). Other pipeline operators interviewed for this study have stated that their systems can safely handle up to 8% hydrogen, but higher concentrations are not recommended at this time. Based on existing systems and owner recommendations, we would state that the blends up to 12% are suitable for use with low yield strength steels, otherwise 8% is the upper limit for most systems (which may include higher yield strength steels). The precise point at which the hydrogen concentration becomes unsuitable for a particular grade of API steel due to embrittlement is not well characterized, especially for pressures characteristic of NG pipelines.

Carbon Dioxide

A number of studies have acknowledged that high purity dry CO_2 whether in gaseous, liquid, or supercritical state is not highly corrosive to pipeline steels (Barrie et al. 2005; Buit et al. 2011; Colahan et al. 2021; Glass 2019; Morland and Dugstad 2016; Morland and Svenningsen 2021; Zeng et al. 2016). The primary risk for metallic materials associated with transporting supercritical CO_2 (sCO_2) are impurities present in the pipeline or in the flowing gas. It is recognized that water is soluble in sCO_2 at very low concentrations (around 500 ppm), but there is disagreement on the precise value (Buit et al. 2011; Morland and Svenningsen 2021). As a result, any appreciable moisture present in pipelines carrying sCO_2 is likely to result in condensation of moisture in the lowest and/or coolest sections of the pipeline with subsequent dissolution of CO_2 in the condensate. This results in formation of carbonic acid which is highly corrosive to pipeline grade steels. Likewise, the presence of other impurities, such as H_2S and SO_2, can result in formation of other acids that are very aggressive toward pipeline steels. The presence of air is also detrimental as even small levels of O_2 may contribute to aggressive corrosion (Rosli et al. 2016; Morland et al. 2019). It has been observed that increasing the flow rate can also enhance the corrosion rate of the exposed steels (Morland and Svenningsen 2021).

A large number of studies have investigated the impacts of impurities which are typically present in CO_2 generation and transport (Aursand et al. 2014; Brown et al. 2017; Choi et al. 2017; Dugstad et al. 2011, 2013; Okezue and Kuvshinov 2017; Paul 2017; Raben et al. 2017; Rosli et al. 2016; Skaugen et al. 2016; Wang et al. 2011; Wetenhall et al. 2014; Witkowski et al. 2013; Zhang et al. 2018). These impurities can be removed, but the cost of such cleaning would likely be greater than the benefits of transporting CO_2 through pipelines. Preventive methods (e.g., coatings) have also been considered, but their application would necessitate careful inspection to ensure performance (Choi et al. 2017; Paul 2017).

It is important to keep in mind that the high pressures (>7.4 MPa, or 1,070 psi) necessary to maintain sCO_2 exceed the design limits of many transmission pipelines. The high pressure and supercritical state of CO_2 may also accelerate crack propagation, which subsequently may require thicker pipe walls than are in current use (Aursand et al. 2014).

Ammonia

To overcome some of the energy and safety concerns associated with transporting and storing of hydrogen, ammonia was also evaluated as an alternative source of hydrogen. These concerns include safety (hydrogen is easily ignited and explosive), compatibility (resulting from hydrogen metal embrittlement and polymer permeation), and low volumetric energy density (Craig 1990; Davies 2006; Schweitzer 1986). Because of these issues, ammonia is being considered as a hydrogen carrier as it is more compatible with metals and plastics and will not ignite or explode during a sudden release (Aziz et al. 2020; Schweitzer 1986). Liquid hydrogen requires cryogenic conditions to achieve −253°C, and has a relatively low energy density (by volume) of 8.5 MJ/liter (MacFarlane et al. 2020). In contrast, ammonia can be stored as a non-cryogenic liquid at moderate pressure (0.9 MPa or 9 bar) and has an energy density of 12.7 MJ/liter. As result, ammonia is a safer, more energy-dense hydrogen carrier than hydrogen alone. These advantages (safety, compatibility, energy density, reduced cooling needs) have made ammonia an attractive alternative to pure hydrogen, in spite of the energy penalties associated with ammonia formation and dissociation. Since ammonia is currently utilized in large-scale volumes as an agricultural fertilizer, refrigerant gas, and in the manufacture of explosives, pesticides and other chemicals, the infrastructure to produce, store, and utilize ammonia is well established and provides a significant amount of information on the compatibility of ammonia with candidate structural containment materials (Aziz et al. 2020).

As one of the largest chemicals produced and transported worldwide, the compatibility properties of ammonia (both anhydrous and aqueous) are well known for most infrastructure metals and polymers. Ammonia is not easily ignited but will burn in air if it reaches 650 °C, which is well above its storage and transport conditions. There are a number of published assessments of the corrosivity of ammonia toward structural metallic materials used for natural gas pipelines (Craig 1990; Davies 2006; Schweitzer 1986). These assessments show that at typical transport pipeline temperatures, carbon and 316 stainless steels have excellent corrosion resistance (<.051 mm/year, or <0.002 inches/year) in anhydrous ammonia. However, carbon and low-alloy steels are susceptible to stress corrosion cracking if certain contaminants, particularly oxygen, are present in the ammonia. Interestingly, the addition of a small amount of water to the ammonia (0.2% is normally used) is known to inhibit the cracking mechanism (Davies 2006). Copper-containing alloys, i.e., brass, bronze and copper, are generally suitable for high purity anhydrous ammonia service, but these alloys are susceptible to stress corrosion cracking if low levels of impurities like H_2O are present in the ammonia; consequently, these alloys are not generally recommended for service with ammonia.

Polymer Pipeline Materials

Polymer materials are employed throughout the natural gas transport and distribution systems. Transmission pipes are primarily composed of metals, such as cast iron and steel. The high pressures required to move NG in transmission systems preclude the use of polymers as structural components, but inner epoxy liners or coatings are common. However, the lower pressures employed in distribution lines enable the use of plastics as structural piping. The earliest plastic pipes for natural gas distribution include polyvinyl chloride (PVC), which has now been replaced by polyethylene. Plastic piping is becoming more predominant in distribution systems because it is less expensive and easier to install than metals. Nylon is also being considered as a NG pipe material, but it has yet to be approved. To date, there are two polymer materials extensively used in natural gas piping: polyethylene and epoxy resins. Polyethylene is classified as thermoplastic material (softens and can be shaped at elevated temperatures) while epoxies are thermosetting plastics (do not soften at elevated temperatures). A wider range of polymer types are used in compressor stations. These include elastomers, which can be easily deformed under stress, but return to most of their original shape when the stress is removed. In contrast to metals, structural polymers are chemically inert to most acids. However, their less dense nature and inherent porosity (in the case for elastomers) mean that they have much higher permeability to gases than metals.

Over the past 30 years, polyethylene (PE) pipe has become the primary pipe material for gas utility distribution systems operating below 0.7 MPa (100 psi). There are over 500,000 mi of distribution main plastic piping in the United States. This pipe material is flexible, corrosion resistant, and inexpensive (AGA 2022). Both epoxies and polyethylene are considered suitable for use with ammonia.

However, the performance of epoxy coatings utilized in steel transmission pipelines with sCO_2 is not known. Investigations into polymer compatibility with sCO_2 have focused primarily on thermoplastic and elastomer materials used as seals and gaskets. Little research has been conducted on investigating thermosetting resin (epoxy) performance with sCO_2. Low levels of dissolution, or swell, may cause the epoxy coating to spall and be carried along by the sCO_2 flow. Once in the flow stream, the spalled epoxy chaff will plug filters, foul pumps, and restrict the operation of valves and meters. In addition, any removed coating will provide direct access of the sCO_2 to the pipe metal surface where it may cause corrosion (if there are impurities present). Studies are needed to ensure that the epoxy resins used as inner coatings in NG transmission pipelines are compatible with sCO_2 under typical operating conditions before introducing sCO_2 in these systems.

Natural Gas Compressor Station Components and Materials

This effort attempted to identify the components and materials of their construction for compressor and regulator stations. This survey was predominantly restricted to components and materials listed by current manufacturers on their open product specification guides. With notable exceptions, details pertaining to specific material grades and/or tradenames were not provided. While it is probable the survey did not capture the full range of component materials, it does show, that for the many companies surveyed, there is commonality of the material type, irrespective of the manufacturer or supplier. An artifact of this approach is that legacy components and materials may be excluded from the tables since current product literature does not include legacy products.

Metallic Materials

Components of compressor stations contain various metals and alloys, including aluminum and common steel grades. Copper and its alloys are not especially prone to hydrogen embrittlement and will likely be suitable for use with hydrogen and its blends (Lee 2016). Other steels, including new specialty steels, are also present and their potential for hydrogen embrittlement needs to be determined.

Aluminum has some susceptibility to hydrogen embrittlement, but to a much lesser extent that steel. For Al alloys, hydrogen permeation is more pronounced when moisture (water vapor) is present than for dry conditions. As with steel, hydrogen uptake in aluminum reduces fracture toughness, cyclic loading, breaking stress, and elongation to failure (Lee 2016). The effects on aluminum are less well understood compared to steel and iron, but an undisturbed and resilient oxide surface layer greatly influences hydrogen permeation by restricting hydrogen movement to the subsurface region. Water is believed to alter the surface layer allowing hydrogen atoms to diffuse into the metal interior. In general, however, hydrogen is less soluble in aluminum than steel, but it will accumulate in the grain boundary regions causing crack growth along the boundaries. There is also little information indicating whether zinc is affected by, or will embrittle, with hydrogen.

A listing of compressor station components and the metallic materials of construction are shown in Table 1. This listing is by no means comprehensive and was pulled from product information sheets from a number of manufacturers. It is important to note that carbon steels are also utilized, but the exact grade and chemistry for these steels is not defined. Other metals that are mentioned are stainless steel, aluminum, zinc, and brass. In this chart, compatibility is based on known embrittlement issues and industry recommendations. Other factors in determining compatibility are application stress, temperature, and variability (cycling) in all application and environmental conditions. Solid metal components that are not subjected to stress, such as housings or casings, may be suitable for use with hydrogen. However, highly stressed components, such as diaphragms and springs, will be greatly impacted by hydrogen exposure. Compatibility with hydrogen is assessed for neat hydrogen and low-level (<20%) blends with natural gas. As shown in the table, the majority of the compressor station components are not considered suitable for use with pure hydrogen. Only those containing copper (brass) and stainless steel were considered suitable. Aluminum, steel, and iron were not considered suitable for use based on studies demonstrating hydrogen embrittlement of these materials in neat hydrogen. For the low-level blends, the compatibility performances are uncertain. As seen in the chart, many of the components and materials were labeled as questionable, since there is not enough information on the metal composition to determine suitability. Key areas of concern are the springs and diaphragms. These components are highly stressed and, therefore, the chances for failure are much higher than for unstressed components. As seen in the table, the most common spring material is steel. The compositional range of spring steels is broad. These are typically high strength, highly elastic, medium carbon, steels that also include some grades of stainless steels. However, because

Table 1. Listing of compressor station components, sub-components and metallic materials of construction

Component	Sub-component	Material	100% hydrogen or high blend levels	Low-level hydrogen blends	sCO_2	Ammonia	Comments
Plug valve	Full assembly	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
Ball valve	Entire valve assembly	Brass	Suitable	Suitable	Suitable	Incompatible	—
		301 stainless steel	Suitable	Suitable	Suitable	Suitable	—
		Carbon steel	Incompatible	Questionable	Suitable	Suitable	—
Safety shut valve	Casting	Iron	Incompatible	Questionable	Suitable	Suitable	—
		Steel	Incompatible	Questionable	Suitable	Suitable	—
	Diaphragm casings	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Steel	Incompatible	Questionable	Suitable	Suitable	—
	Diaphragm plates	Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Orifice	Brass	Incompatible	Suitable	Suitable	Incompatible	—
		Steel	Incompatible	Questionable	Suitable	Suitable	—
	Fasteners	Steel	Incompatible	Questionable	Suitable	Suitable	—
	Springs	Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Cap	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
Appliance regulators	Housing	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Brass	Suitable	Suitable	Suitable	Incompatible	—
	Vent emitters	Brass	Suitable	Suitable	Suitable	Incompatible	—

Table 1. (*Continued.*)

Component	Sub-component	Material	100% hydrogen or high blend levels	Low-level hydrogen blends	sCO_2	Ammonia	Comments
Pilot operated regulators	Closing spring	Music wire	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Valve body	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
	Orifice	Brass	Suitable	Suitable	Suitable	Incompatible	—
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Valve stem	Aluminum-plated steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Lever pin	303 stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Lever	Zinc-plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Upper diaphragm plate	Zinc-plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Lower diaphragm plate	Zinc-plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
		Aluminum	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Vent screen	Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Adjustment ferrule	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Cap	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Diaphragm case	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Internal monitor orifice	Brass	Suitable	Suitable	Suitable	Incompatible	—
	Screwed body casting	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
	Flanged body casting	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
		Ductile iron	Incompatible	Questionable	Suitable	Suitable	—
		Steel	Incompatible	Questionable	Suitable	Suitable	—
	Pilot supply line	Copper	Suitable	Suitable	Suitable	Incompatible	—
	Tubing	Copper	Suitable	Suitable	Suitable	Incompatible	—
	Springs	Zinc-plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Main spring case	Ductile iron	Incompatible	Questionable	Suitable	Suitable	—
	External fittings	Brass	Suitable	Suitable	Suitable	Incompatible	—
	Pitot trim	Brass	Suitable	Suitable	Suitable	Incompatible	—
	Pitot internals	Zinc-plated steel	Suitable	Suitable	Suitable	Incompatible	—
		Zinc	Suitable	Suitable	Suitable	Incompatible	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Filter	Brass	Incompatible	Suitable	Suitable	Incompatible	—
Direct acting regulators	Housing	Carbon steel	Incompatible	Questionable	Suitable	Questionable	Not a high stress component
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Valve body	Iron	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
		Ductile iron	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Orifice	Brass	Suitable	Suitable	Suitable	Incompatible	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Valve seat	Brass	Suitable	Suitable	Suitable	Incompatible	Critical component under high stress
	Valve stem	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Bonnet & Cage	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Adjustment screw	Brass	Suitable	Suitable	Suitable	Incompatible	—
	Diaphragm Case	Steel	Incompatible	Questionable	Suitable	Suitable	—
	Internal parts	Brass	Suitable	Suitable	Suitable	Incompatible	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—

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Table 1. (*Continued.*)

Component	Sub-component	Material	100% hydrogen or high blend levels	Low-level hydrogen blends	sCO_2	Ammonia	Comments
Line and service regulators	Housing	Epoxy coated aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Bonnet	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Body casting/valve body	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
		Ductile iron	Incompatible	Questionable	Suitable	Suitable	—
	Orifice	High strength brass	Suitable	Suitable	Suitable	Incompatible	—
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Internal monitor orifice	360 Brass	Suitable	Suitable	Suitable	Incompatible	—
	Diaphragm plates	Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
		Zinc plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
		Aluminum	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Diaphragm case	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Cast iron	Incompatible	Questionable	Suitable	Suitable	—
	Lever	Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
		Zinc plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
	Lever pin	303 Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Vent screen	Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Valve stem/extension	Die cast zinc	Suitable	Suitable	Suitable	Incompatible	Critical component under high stress
		Aluminum	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Valve stem guide	Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Adjustment ferrule	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Fasteners	Steel	Incompatible	Questionable	Suitable	Suitable	—
	Cap	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Springs	Steel	Incompatible	Questionable	Suitable	Suitable	—
	Spring seat	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
High pressure regulators	Housing	Cast iron	Incompatible	Questionable	Suitable	Suitable	—
		Steel	Incompatible	Questionable	Suitable	Suitable	—
	Body/bonnet	Zinc	Suitable	Suitable	Suitable	Incompatible	—
		Cast iron	Incompatible	Questionable	Suitable	Suitable	—
		Ductile steel	Incompatible	Questionable	Suitable	Suitable	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Brass	Suitable	Suitable	Suitable	Incompatible	—
	Diaphragm assembly	Zinc	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Diaphragm plate	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Diaphragm head	Cd plated steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Diaphragm piston	Zinc plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
	Pin	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Regulator spring	Zinc plated steel	Incompatible	Questionable	Suitable	Incompatible	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
		Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Upper ring seat	Zinc	Suitable	Suitable	Suitable	Incompatible	Critical component under high stress
		Steel	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Spring case	Brass	Suitable	Suitable	Suitable	Incompatible	—

Table 1. (*Continued.*)

Component	Sub-component	Material	100% hydrogen or high blend levels	Low-level hydrogen blends	sCO_2	Ammonia	Comments
	Orifice	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Cap	Brass	Incompatible	Suitable	Suitable	Incompatible	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Valve disk	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Valve spring	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Disk holder/stem guide	Stainless steel	Suitable	Suitable	Suitable	Suitable	—
		Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Adapter screws/screws	Steel	Incompatible	Questionable	Suitable	Suitable	—
		Stainless steel	Suitable	Suitable	Suitable	Suitable	—
	Vent screen	Monel	Suitable	Suitable	Suitable	Suitable	—
	Cap	Brass	Suitable	Suitable	Suitable	Incompatible	—
Rotary meter	Case	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Impellers	Aluminum	Incompatible	Questionable	Suitable	Suitable	Critical component under high stress
	Bearings	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Masking plate	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Pressure plate	Stainless steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress
	Cover	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Plugs	Brass	Suitable	Suitable	Suitable	Incompatible	—
	Front bearing plate	Aluminum	Incompatible	Questionable	Suitable	Suitable	—
	Oil slinger	Steel	Suitable	Suitable	Suitable	Suitable	Critical component under high stress

the specific grades and chemistries of these steels are not provided by the manufacturers, it is impossible to definitively state their suitability for use with hydrogen with any precision.

The results in Table 1 indicate that the metallic materials in the compressor station components are suitable for use with CO_2 (supercritical or otherwise). Keep in mind that this is only true if there is no moisture present. The presence of a separate water phase will drive the formation of carbonic acid, which is corrosive to many steels and iron.

The compatibility with ammonia was determined based on known material incompatibilities with copper and copper alloys, especially brass. Copper, brass, and bronze are compatible with anhydrous ammonia, since it does not contain appreciable water. On the other hand, gaseous ammonia that contains water as a significant impurity will be highly corrosive to these metals. As seen in the table, brass is used in a range of applications, including housings, orifices, valve seats, and screws. As such, all brass components are listed as incompatible in the table.

Polymer Materials

Except for the epoxy coatings lining the inner walls of transmission pipes and PE distribution lines, the majority of polymer materials in natural gas systems are used in the compressor stations. The assorted compressors, valves, fittings, and regulators comprising a compressor station possess a variety of materials in contact with NG, especially polymers, which are extensively used as seals and diaphragms. In this study we have not considered the component performance associated with the change in thermo-physical properties of the fluids. For example, flow meters based on sound speed are designed for use with natural gas and will not read properly for hydrogen blends, CO_2, or ammonia. Other factors like the relatively high pressures needed to maintain supercritical CO_2 preclude the use of sCO_2 in plastic piping that is only rated to 0.7 to 3.5 MPa (100 to 500 psi). A quick survey of the literature reveals the following:

1. The majority of hydrogen blend compatibility studies with polymers have been performed under extremely high pressures that are not representative of conditions in natural gas pipelines (Simmons et al. 2021b). These conditions are more representative of fuel cells, which requires operation under high pressures. Another key concern with these studies is the lack of a natural gas baseline for comparison. There is, however, operational experience with hydrogen-NG blends that indicates suitability with existing infrastructure elastomers and plastic materials.
2. The transport efficiency of CO_2 is greatly facilitated using dense or supercritical CO_2. As a result, compatibility studies with polymers and pipeline materials have been primarily concerned with CO_2 in the supercritical state. The results of these studies have shown that many infrastructure elastomers and some plastics are not compatible with sCO_2 (Ansaloni et al. 2020; Menon et al. 2019). It is important to note that these studies have focused on polymers that are currently in use in sCO_2 pipelines and have not considered those polymers currently in use in NG systems. Low density, gaseous CO_2 (either neat or as a blend with NG) has not been subjected to significant research, but it should be compatible with most infrastructure polymeric materials (depending on blend level).
3. Ammonia compatibility with polymer materials is well understood. Ammonia pipelines exist for the transport of ammonia for agriculture use and there is considerable experience with ammonia handling and transport. The compatibility of ammonia with polymers varies greatly with polymer type.

Failure criteria for polymers are typically based on swell and permeability. Property changes often accompany swelling, which include a reduction in strength, modulus, and hardness. In turn, these property changes affect the compression set of polymer seals and sealing ability. Structural changes associated with crosslinking, chemical bonding and composition (such as the ingress of fluid and the extraction of additive components) will potentially alter the glass transition temperature making the polymer more brittle or rubbery during operation, which may be beyond their design application limits (Menon et al. 2019). Failure criteria associated with the glass to rubber transition temperature is well defined. However, swell and other property changes need to be correlated to performance specifications of the component, which necessitates field-based or component-level testing. Another area of uncertainty is leakage due to permeation. To date, there are no available permeability limits defined for hydrogen and its blend with NG, carbon dioxide, or ammonia in current pipelines.

A listing of the compressor station components and their polymer sub-components is shown in Table 2. The information provided in this table was derived from product specification sheets from the following manufacturers: Marshall Exelsior, CVS, Maxitrol, NMT (Norgas Meter Technologies), Bryan Donkin, Itron, Richards, Honeywell, and Romet. It is worth noting that the materials of construction can, and do, vary by manufacturer. As a result, the polymer materials listed in the table for a given component type (i.e., o-ring) are inclusive from several manufacturers.

As seen in the table, polymers are primarily used as seals, diaphragms, o-rings, and valve seats. The listed o-ring materials are elastomers including acrylonitrile butadiene rubbers (NBRs), hydrogenated NBR (HNBR), and fluorocarbon (FKM or Viton™). Seals and gasket materials are primarily elastomers, such as NBR, HNBR, FKM, polyurethane, and perfluoroelastomer (FFKM or Aflas™), but polytetrafluoroethylene (PTFE or Teflon™) is used in some applications. The material listing for valve seats and heads is broad and includes additional plastic materials such as acetal and nylon, as well as silicone. For each of these polymer materials, their compatibilities were assessed for use with hydrogen (and blends with NG), sCO_2, and ammonia. The performance of the component, whether suitable or incompatible, was determined by the compatibility of any one of the subassembly polymers with the exposed gas (or liquid).

Hydrogen Blends with Natural Gas

As can be seen in Table 2, the listed compressor station components are expected to be suitable for use with hydrogen (and its blends with NG). This finding confirms the experience of others who have been producing synthetic natural gas with a high hydrogen content (10%–12%) (Hawai'iGas 2022). For polymers, the biggest area of risk will be the compressor station. Epoxies are known to be highly chemically resistant as is polyethylene. The US Department of Energy HyBlend Program and the Hydrogen Materials Compatibility Consortium have been conducting studies evaluating the compatibility of hydrogen with pipeline materials, including polymers (HYBLEND 2021). As stated previously, most of these studies were conducted under extremely high pressures associated with fuel cell use. The DOE H-Mat program evaluated NBR and ethylene propylene diene monomer (EPDM) at 100 MPa (14,000 psi) of hydrogen (Simmons et al. 2021b). This study showed a decrease in

Table 2. Listing of compressor station components, sub-components and polymeric materials of construction

Component	Sub-component	Material	Hydrogen blend	sCO_2	Ammonia	Comments
Ball valve	Valve seat	PTFE (Teflon™)	Suitable	Incompatible	Suitable	—
	Stem o-ring	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Stem packing	PTFE (Teflon™)	Suitable	Incompatible	Suitable	—
Safety shut valve	Diaphragm	NBR with nylon reinforcement	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve heat/seat	Polyurethane	Suitable	Incompatible	Incompatible	—
	O-rings	NBR	Suitable	Incompatible	Questionable	—
	Seal	Graphite	Suitable	Suitable	Suitable	Not a polymer
Appliance regulators	Diaphragm and valve seat	NBR with and without nylon reinforcement	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve head	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Cap and assembly	Molded plastic	Suitable	Suitable	Suitable	Plastic was not specified, but should be ok based on survey
	O-rings	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve seat	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Silicone	Suitable	Suitable	Questionable	Ammonia suitability depends on gas temperature
	Lower diaphragm plate	Polyester	Suitable	Suitable	Suitable	—
	Diaphragm	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Nylon	Suitable	Suitable	Suitable	—
	Vent valve/seat	Neoprene	Suitable	Incompatible	Suitable	—
		Acetal/NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Adjustment Ferrule	Acetal	Suitable	Suitable	Suitable	—
	Seal cap	Polyethylene	Suitable	Suitable	Suitable	—
Pilot operated regulators	Valve seat	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Silicone	Suitable	Suitable	Questionable	Ammonia suitability depends on gas temperature

Table 2. (*Continued.*)

Component	Sub-component	Material	Hydrogen blend	sCO_2	Ammonia	Comments
	Diaphragm	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Nylon	Suitable	Suitable	Suitable	—
	Vent valve/seat	Neoprene	Suitable	Incompatible	Suitable	—
	Pilot trim	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
Direct acting regulators	Diaphragm	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Nylon	Suitable	Suitable	Suitable	—
	Valve disk	Neoprene	Suitable	Incompatible	Suitable	—
		NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	O-rings	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Sealing washer	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve seat	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve stem bushings	Nylon	Suitable	Suitable	Suitable	—
	Seals	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		FKM (Viton™)	Suitable	Incompatible	Incompatible	—
High pressure regulators	Diaphragm	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Neoprene	Suitable	Incompatible	Suitable	—
	Diaphragm assembly	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		Neoprene	Suitable	Incompatible	Suitable	—
		FKM (Viton™)	Suitable	Incompatible	Incompatible	—
		PTFE (Teflon™)	Suitable	Incompatible	Suitable	Used specifically for use with ammonia
	Seat	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve disk assembly	Nylon	Suitable	Suitable	Suitable	—
	Disk holder	Nylon (2,000 psi)	Suitable	Suitable	Suitable	—
		NBR (1,000 psi)	Suitable	Incompatible	Questionable	Depends on the NBR grade
	O-rings	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Gaskets	Nylon valve disk + neoprene gaskets (−20 to 180 F)	Suitable	Incompatible	Suitable	—
		PTFE (Teflon™) valve disk + FKM (Viton™) gasket	Suitable	Incompatible	Incompatible	—
		PTFE (Teflon™) valve disk + EPDM gasket	Suitable	Suitable	Suitable	—
Backpressure regulators	Seat disk	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Seal	HNBR	Suitable	Suitable	Incompatible	—
		FKM (Viton™)	Suitable	Incompatible	Incompatible	—
		NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		FFKM perfluorocarbon	Suitable	Suitable	Suitable	—
	Diaphragm	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Gaskets	Neoprene	Suitable	Incompatible	Suitable	—
	O-rings	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
	Valve head/seat	Polyurethane	Suitable	Questionable	Incompatible	Allorings.com
	Valve plug	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
Honeywell pressure regulators	Valve spacer, top cap, and valve	Acetal	Suitable	Suitable	Suitable	—
Rotary meter	O-ring	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
High pressure control valves	Seals	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		FFKM perfluorocarbon (Aflas™)	Suitable	Suitable	Suitable	—
		FKM (Viton™)	Suitable	Incompatible	Incompatible	—
Temperature controllers	Seals	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		HNBR	Suitable	Suitable	Incompatible	—
		FKM	Suitable	Incompatible	Incompatible	—
		FFKM perfluorocarbon (Aflas™)	Suitable	Suitable	Suitable	—
		Polyurethane	Suitable	Questionable	Incompatible	Allorings.com
		PTFE (Teflon™)	Suitable	Incompatible	Suitable	—
Pilots and relays	Seals	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		HNBR	Suitable	Suitable	Incompatible	—
		FKM	Suitable	Incompatible	Incompatible	—
	Diaphragm	NBR	Suitable	Incompatible	Questionable	Depends on the NBR grade
		FKM	Suitable	Incompatible	Incompatible	—
	O-rings	NBR	Suitable	Suitable	Questionable	Depends on the NBR grade
		HNBR	Suitable	Suitable	Incompatible	—
		FKM	Suitable	Incompatible	Incompatible	—

storage modulus for EPDM with fillers and plasticizer additives. NBR also exhibited significant compression set. In practice, these extremely high-test pressures are not representative of NG pipeline conditions, which rarely exceed 10.3 MPa (1,500 psi). Therefore, much of this information is only of limited use for NG pipeline applications. Another study by Yamabe (Yamabe and Nishimura 2009) also examined the effect of hydrogen on EPDM and NBR at high pressure. Blister damage due to rapid decompression was observed for NBR with high filler (additive) loadings.

More realistic (low pressure) studies have been conducted with pure hydrogen. One such study done by French researchers showed that nylon and polyethylene were not structurally degraded by hydrogen under relevant pressures (Castagnet et al. 2012). In this study, tensile tests were performed in hydrogen at 3 MPa (435 psi) and no differences in mechanical properties or structural changes were found when compared to running tests in air at ambient conditions. Structural testing of polyethylene samples exposed to hydrogen gas at pressures less than 1.7 MPa (250 psi) was conducted by PNNL (Simmons et al. 2021a). The exposed samples experienced a slight reduction in both crystallinity and hardness, but were otherwise unaffected. This effect was found to be temporary, and the samples returned to their original property values when the hydrogen was removed. To date, we have not found any low-pressure study examining hydrogen blends with natural gas.

Supercritical, or High-Density, CO_2

A number of studies have examined the compatibility of polymers with supercritical, or high density, CO_2 (Menon et al. 2019). Information from these studies was compiled and further examined for compatibility assessment. Liquid CO_2 is known to be a good solvent for many elastomers and thermoplastics since it readily forms weak bonds with many polymers. Polymers particularly susceptible include fluorocarbons (FKMs) and acrylonitrile butadiene rubbers (NBRs), which are two of the most widely used o-ring and seal materials. The absorption of CO_2 increases the swell and softening of these polymers. This effect is most pronounced for supercritical CO_2 (Ansaloni et al. 2020). Low-pressure gaseous CO_2 will exhibit similar trends, but at a significantly reduced level, and is compatible with many elastomer materials based on existing polymer manufacturer compatibility tables.

The results in Table 2 show that sCO_2 will not be suitable with current compressor station components (with the possible exception of Honeywell regulators). This means that alternative compressor and regulator components and protocols will be needed to transport sCO_2 in NG pipelines. Gaseous CO_2, either as a blend with NG or neat, will have improved compatibility with polymers, especially at the lower pressures typical of transmission and distribution lines.

Ammonia

For ammonia, the results are more mixed. Ammonia has excellent compatibility with many infrastructure polymers. It is important to note that any piping system used to transport ammonia must have ammonia-rated materials and components (including hosing and flexible lines) [MDA Factsheet (MDA 2022)]. There are a number of standards related to ammonia compatibility that have to be met; they include ANSI B31.3 and ANSI K61.1. However, these standards are not applied to NG pipelines. Polyethylene and epoxies are suitable for use with ammonia; therefore, the ammonia transport through existing NG pipeline sections is feasible. However, as can be seen in Table 2, many compressor station components are not suitable for use with ammonia. For elastomers, materials not acceptable for use with ammonia include fluorocarbons, PTFE, and SBR. Other questionable polymers are polyurethane, epichlorohydrin rubber (ECO) and fluorosilicone. Suitable elastomers include some, but not all, NBRs, EPDM, neoprene or CR, and FFKM). Ammonia is also highly incompatible with polytetrafluoroethylene (PTFE) and nylons which are standard seal and diaphragm materials. Its compatibility with NBR depends on the grade and application.

Conclusions

A review of material performance literature and a comprehensive listing of the materials comprising natural gas piping systems was conducted to evaluate whether existing natural gas piping systems could be repurposed for transporting hydrogen blends with natural gas, ammonia, and CO_2. For API pipeline steels, those grades that are X52 and less appear to be suitable for use with low blend levels (<15% hydrogen) based on embrittlement studies and existing blend histories. The exact blend level of suitability will depend on the existing pipeline steel grades. The existing pipes would be suitable for handling ammonia and gaseous CO_2; however, supercritical CO_2 would not likely be suitable due to the pressure requirements exceeding the piping strength limits. Supercritical CO_2 is a known solvent for many polymers and its impact with epoxies and other thermosetting resins has not been seriously studied. Therefore, before sCO_2 is introduced in a NG pipeline, its compatibility with these epoxy coatings must be determined.

The compressor and deregulator stations are particularly vulnerable since they contain a wide array of metals and polymers. The uncertainty surrounding many of the steel grades and aluminum alloys would limit the blend level of hydrogen in natural gas. Additional research is required to evaluate alloy compatibility for hydrogen blends of greater than 15% over the expected operating pressure range. Many of the compressor station component materials are also unsuitable for use with sCO_2 and ammonia. Any scenario considering moving sCO_2 or ammonia would need to bypass existing compressor and regulator stations in order to be feasible. The gases most suitable for repurposing are low hydrogen blends and gaseous CO_2. However, there are still many unknowns regarding the impacts these gas chemistries have on compressor station polymers and impurity effects. Therefore, more study is recommended for these two gas streams.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

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